

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

ARCHITECTURAL DIAGRAM

Visualization Tool for Electric Vehicle Charge and Range Analysis

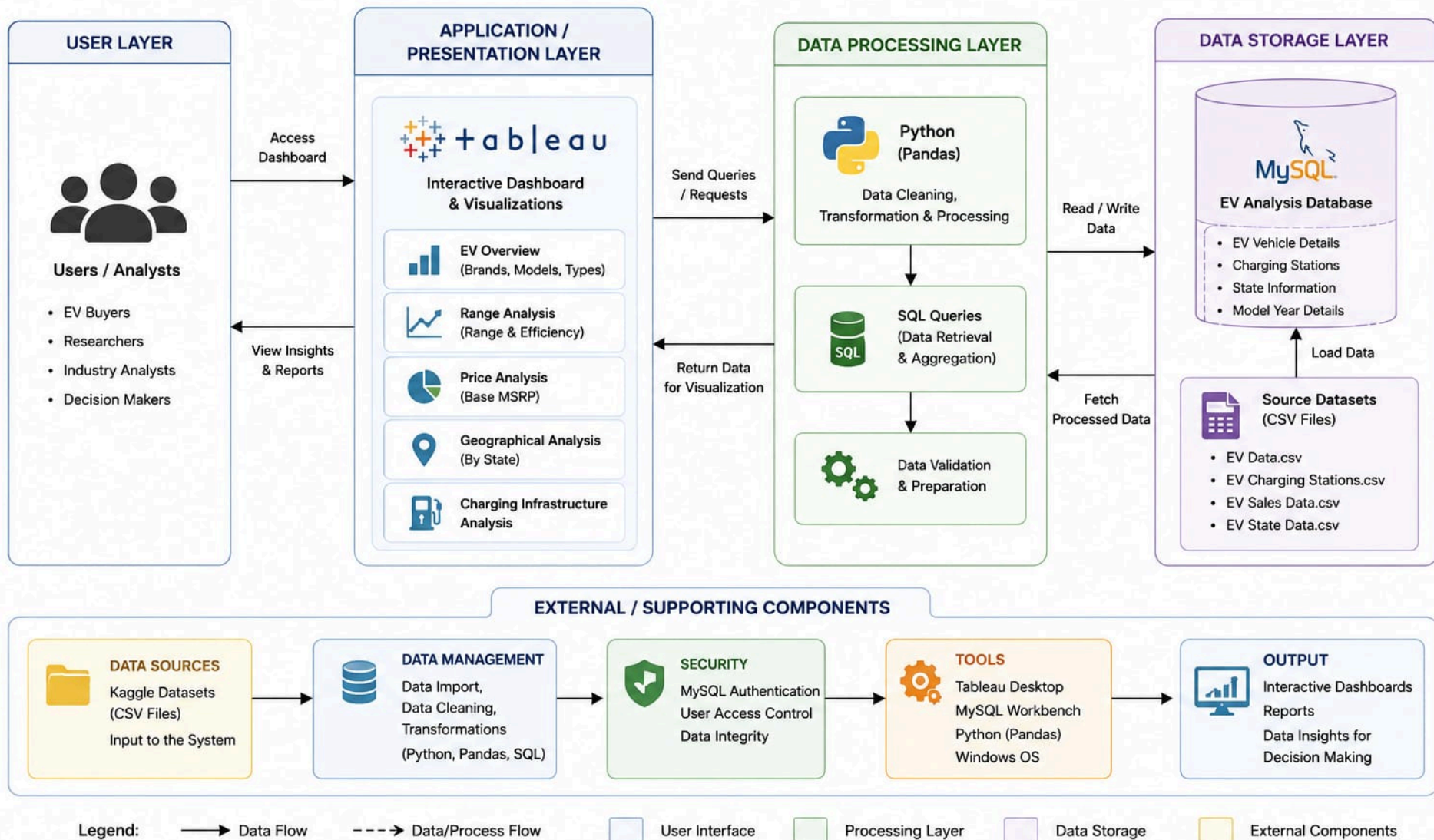


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for EV analysis	Tableau
2.	Application Logic-1	Data cleaning and preprocessing	Python (Pandas)
3.	Application Logic-2	SQL queries and data management	MySQL
4.	Application Logic-3	Data Visualizations and Dashboard Generation	Tableau
5.	Database	Stores EV datasets	MySQL
6.	Cloud Database	Not Used	N/A
7.	File Storage	Stores CSV datasets	Local File System
8.	External API-1	Not Used	N/A
9.	External API-2	Not Used	N/A
10.	Machine Learning Model	Not Used	N/A
11.	Infrastructure (Server / Cloud)	Local deployment	Windows PC/ Local System

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Libraries used for data processing	Python, Pandas
2.	Security Implementations	Database authentication and authorized access	MySQL Authentication
3.	Scalable Architecture	Additional EV datasets can be added easily	MySQL + Tableau
4.	Availability	Dashboard is available whenever the local system is running	Tableau Desktop
5.	Performance	Fast data retrieval using SQL queries and optimized dashboards	MySQL, Tableau

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>